

The **posim** Graphical Scene Window

Research, Design & Complete Guide (`scene_info`)

physical_object_simulator

July 22, 2026

This is the typeset twin of `scene_info.md` — companion to `grammar.pdf` and `physical_object_simulator.pdf`.

This document explains, for a reader with **zero prior knowledge**:

1. what the graphical scene window is and how to drive it,
2. the **research** performed on seven established physics simulators before designing it,
3. a **comparison chart and concept tree** of that “scene science”,
4. which ideas we recommend (and chose) to **port to pure Rust**, and
5. the full technical documentation of the implementation, its asynchronous notebook link, and how every claim was **verified**.

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1 What is the scene window?

`posim` is a physics simulator you drive by typing commands (`NEW SPHERE { ... }`, `STEP 0.1`, ...). Numbers are exact but hard to *picture*. The **scene window** is a live 3-D view of every object in your simulation that opens **in your web browser, outside the notebook**, the moment you type:

```
scene create
```

You get back something like:

```
scene window created: http://127.0.0.1:41234/
(opened in your browser; if no window appeared, open that address yourself)
showing 4 entities; SCENE START begins the evolution - HELP lists all scene commands
```

The window shows **all simulator entities** (points, spheres, cuboids, tori, disks, cylinders, dumbbells), a ground grid, the world axes, motion trails, and object labels (an object registered with `NEW ... AS <name>` is labeled by that name). If the notebook created a rigid bounding box (`BOX <size>`), the window draws its interior as a dashed light-blue wireframe — the six infinitely-massive wall slabs that implement it are represented by that wireframe, never drawn as six giant cuboids. The window keeps itself up to date over a private network link to the simulator — you never copy data by hand. It can also *talk back*: errors and user actions inside the window flow to the notebook asynchronously (§7).

No extra software is needed: the whole server is built into `posim` itself using only the Rust standard library — zero external dependencies, zero `unsafe` code.

1.1 The toolbar (top of the window)

control	what it does
Start	begin time-stepped evolution, forward in time
Pause	freeze the evolution (resume with Start)
Stop	halt evolution <i>and clear the recorded history</i>
Reverse	play backward through the recorded history
Reset	re-initialize the playback: every value and the time return to their initial values ; Start then runs the simulation again from the beginning
step back / step	single step backward / forward
dt ____ Set	change the playback time step
zoom + / zoom -	zoom in / out
View	reset the camera to its home position
Grid / Trails / Labels	toggle the ground grid, motion trails, object labels
Contacts	toggle contact-normal arrows drawn at collision points
?	show the controls cheat-sheet

1.2 The “conserved quantities” readout (top-left)

A permanent labeled overlay (just below the toolbar) shows the three conserved quantities of the playback copy, updated live at every frame: **E** (total energy), **P** (total linear momentum) and **L** (total angular momentum about the origin). **P** and **L** are printed as their three components plus the magnitude $|\cdot|$. Watching **E**, **P** and **L** sit still through a collision *is* the physics lesson — the values come straight from the frame protocol’s **energy**, **p** and **l** fields (§5.1), computed on the playback copy every tick.

1.3 The status bar (bottom of the window)

A live one-line dashboard: **connection dot** (green = connected to posim), playback **mode** (stopped / running / paused / reversing), simulated time **t**, time step **dt**, total energy **E**, **bodies** count, **hidden** count, **contacts** count (of the latest playback step), **history** (frames available for Reverse), the **camera** yaw/pitch/distance, and the rendering **fps**.

1.4 Controls (mouse & keyboard) — all verified in a real browser

gesture	effect
left / right arrow keys	translate the view left / right
up / down arrow keys	translate the view up / down
left-click + drag	rotate (orbit) around the scene
mouse wheel	zoom in / out
+ / - keys	zoom in / out (documented keyboard shortcut)
Shift+drag or right-drag	translate (pan) with the mouse
Space	start / pause playback
R	reset the view
G / T / L	toggle grid / trails / labels
C	toggle contact-normal arrows
H or ?	toggle the help overlay

2 The research: seven simulators investigated

Before writing a line of code we sent research agents to study how seven established, widely-used simulators present and control a graphical scene. A summary of each, then the comparison chart.

2.1 PyChrono / Project Chrono (C++, Python bindings)

Multi-physics engine used in engineering and robotics. Rendering is **in-process** through an abstract *visual system* layer with pluggable backends — Irrlicht (`ChVisualSystemIrrlicht`), VulkanSceneGraph (`ChVisualSystemVSG`), raw OpenGL, plus offline Blender/POVRay export. Typical loop: `AttachSystem(sys)`, `Initialize()`, then `Run()/BeginScene()/Render()/EndScene()` around `ChSystem::DoStepDynamics(dt)` — *you* own the loop and the time step; pause/step are gates you place around `DoStepDynamics`. Scene decoration: `AddCamera(pos, target)`, `AddTypicalLights()`, `AddSkyBox()`, `EnableShadows()`. Viewer: left-drag rotate, right-drag pan, wheel zoom, i info panel, Space pause.

2.2 PyBullet (C++ engine, Python module)

The gold standard in robotics labs. **Client-server by design**: a client sends *command structs*, the physics server returns *status structs* — identical code whether the transport is same-process (`connect(DIRECT)/connect(GUI)`), shared memory, UDP, or TCP. Key calls: `stepSimulation()`, `setTimeStep(dt)`, `setRealTimeSimulation(0/1)`, `resetDebugVisualizerCamera(distance, yaw, pitch, target)`, `configureDebugVisualizer(flag, on)`, `addUserDebugLine/Text/Parameter`, `getKeyboardEvents()` / `getMouseEvents()`. The GUI has dockable parameter panels and preview tiles. No built-in reverse.

2.3 SOFA (C++, Qt/OpenGL GUI)

Interactive multi-physics (deformables, biomechanics). `runSofa` is an **in-process Qt app**; the default viewport is `libQGLViewer (QtGLViewer)`. Scenes are declarative trees (`.scn` XML or Python). GUI: **Animate, Step, Reset Scene** buttons and an editable `dt` field; FPS + elapsed time bottom-left; Save/Reset View. Mouse: left rotate, right pan, wheel zoom, Shift+click pokes the simulation. Shortcuts: S screenshot, T ortho/perspective, V video. No reverse.

2.4 Rapier (Dimforge, pure Rust)

Modern 2-D/3-D physics engines in Rust. The core is **fully rendering-agnostic**: `PhysicsPipeline::step()` advances `RigidBodySet/ColliderSet`; you read poses back through handles. The testbed (Bevy + egui) contributes the clearest control vocabulary: `RunMode {Running, Stop, Step}`, display bit-flags (`TestbedStateFlags: WIREFRAME, AABBS, CONTACT_POINTS, ...`), one-shot `TestbedActionFlags` including `TAKE_SNAPSHOT / RESTORE_SNAPSHOT` (serialize the whole world — the germ of *reverse*) and `FRAME_SCENE` (auto-fit camera). Its `GraphicsManager` keeps hash maps from physics handles to render nodes so geometry is sent once and only poses update.

2.5 Open Source Physics — OSP / EJS (Java, Davidson College)

The classic *educational* library. `DrawingPanel` owns the pixel↔world transform with built-in pan/zoom; `PlottingPanel/Dataset` add live graphs; `DataTable` numeric readouts. `AbstractSimulation` runs an animation thread calling `doStep()` $\sim 10\times/s$, with a `SimulationControl` GUI of **Start/Stop**,

Step, **Reset**, **Initialize** buttons, an editable parameter table and a message area; **stepsPerDisplay** decouples physics rate from display rate. Numerics live behind an **ODE/ODESolver** interface (Euler, Verlet, RK4, RK45, ...). EJS generates Java *or* JavaScript from the same model.

2.6 SIMple Physics (Rust, educational)

High-school-oriented Rust simulators (SIMple Mechanics, SIMple Gravity) built on ggez + specs ECS + imgui + Lua scripting. Editable globals (**GRAVITY**, **PAUSED**, **DT**), per-object property panels (right-click), keyboard object creation, Space pause, **graph object properties and export to CSV** for lab reports, Lua scene presets. Its lesson: make every quantity inspectable and tweakable.

2.7 PhysicsHub (web, student-built)

Two related projects: a React/Next.js canvas app (physicshub.github.io) and the earlier Node/p5.js “The Physics Hub”. The p5 version **layers canvases**: **bgCanvas** (chrome + menus), **simCanvas** (the simulation), **plotCanvas** (toggleable live plots), plus grid overlays; sliders/buttons via p5.gui; theory text with MathJax beside every simulation. The browser’s **requestAnimationFrame** is the render loop; widgets mutate parameters read by the next frame.

2.8 The comparison chart

(Split across two tables to stay readable; same columns.)

	PyChrono		PyBullet	SOFA		Rapier
Language / stack	C++ Python)	(+SWIG	C++ engine, Python API	C++ core, Python scenes		Rust
Rendering	Irrlicht / OpenGL	VSG /	OpenGL debug ren- derer	Qt libQGLViewer	+	Bevy/wgpu testbed, WASM demos
Process model	in-process		client-server (shm / UDP / TCP)	in-process		in-process (lib head- less)
Scene / camera API	ChVisualSystem* , AddCamera		resetDebugVisualizer , configureDebugVisual	Scene-graph panel, Save/Reset View		GraphicsManager maps, FRAME_SCENE
Time stepping	DoStepDynamics(dt) user loop		stepSimulation() , setTimeStep	Animate / Step / dt field		RunMode {Running, Stop, Step}
Pause / step / reverse	gate the loop / —		call-when-you-want / —	buttons / —		RunMode / snap- shot+restore
Mouse convention	L-rotate, wheel	R-pan,	L-orbit, wheel, pan	L-rotate, wheel	R-pan,	orbit / pan / zoom + pick
Toolbar / HUD	info panel (i)		sliders, preview tiles	Animate FPS+time	bar,	egui panel, stats flags
Async sim↔GUI	none		command→status mailbox	none		ECS systems

	OSP / EJS	SIMple Physics	PhysicsHub
Language / stack	Java (EJS→JS too)	Rust (ggez, specs, Lua)	JS (React/Next or p5.js)
Rendering	Swing DrawingPanel	ggez 2-D	HTML <canvas>
Process model	in-process	in-process	in-browser
Scene / camera API	DrawingPanel pan/zoom, InteractivePanel	imgui panels, Lua globals	per-sim canvas + sliders
Time stepping	doStep() thread, stepsPerDisplay	update() each frame, DT global	requestAnimationFrame
Pause / step / reverse	Start-Stop-Step-Reset / —	Space pause / —	play-pause-reset / —
Mouse convention	drag interactives	L-drag move, R-click edit	widget-based
Toolbar / HUD	buttons + param table + messages	imgui property panels	sliders, plots, theory panes
Async sim↔GUI	animation thread	shared ECS state	same-thread JS

2.9 The “scene science” concept tree

Which simulator taught us each concept (→ = adopted in posim):

```

scene science
|-- scene lifecycle
|   |-- create / destroy a window          <- PyBullet connect()/disconnect() ->
|   SCENE CREATE / CLOSE
|   |-- describe geometry once, poses often <- Rapier GraphicsManager          -> init
|   vs frame messages
|   |-- declarative scene description       <- SOFA .scn / Bullet URDF          -> init
|   entity list (JSON)
|   '-- refresh / redraw on demand         <- SOFA Reset Scene                ->
|   SCENE REFRESH / REDRAW
|-- camera control
|   |-- orbit camera (yaw,pitch,dist,target)<- PyBullet resetDebugVisualizerCamera ->
|   SCENE ROTATE / server camera
|   |-- translate / pan                    <- Irrlicht & QGLViewer right-drag ->
|   SCENE TRANSLATE, arrows, shift-drag
|   |-- zoom (wheel + keys)                <- all seven                      ->
|   SCENE ZOOM IN/OUT/<f>, wheel, +/-
|   '-- frame-the-scene auto-fit           <- Rapier FRAME_SCENE              ->
|   fit_distance() at create/refresh
|-- playback control
|   |-- run-mode state machine              <- Rapier RunMode                    ->
|   Stopped/Running/Paused/Reversing
|   |-- start / stop / pause / single-step <- OSP AnimationControl              ->
|   SCENE START/STOP/PAUSE, steps
|   |-- reverse via snapshot history        <- Rapier TAKE/RESTORE_SNAPSHOT      ->
|   SCENE REVERSE (ring buffer)
|   '-- settable time step                 <- PyBullet setTimeStep / SOFA dt    ->
|   SCENE SET_TIME_STEP
|-- asynchronous messaging
|   |-- command -> status protocol          <- PyBullet client-server            ->
|   WebSocket JSON both ways
|   |-- window -> notebook events          <- PyBullet debug/user events        ->

```

```

    SCENE EVENTS + {"event":...}
|  '-- transport-agnostic message structs  <- PyBullet shm/UDP/TCP      -> same
    JSON, any front end
'-- educational UX
    |-- toolbar of big obvious buttons      <- OSP SimulationControl    -> the
    toolbar
    |-- status readouts (t, E, fps)         <- SOFA FPS/time, OSP DataTable -> the
    status bar
    |-- trails / plots / layered canvas     <- PhysicsHub plotCanvas      -> grid
    + trails + bodies layers
    |-- hide/show and visual flags          <- Bullet configureDebugVisualizer ->
    SCENE HIDE / SHOW
    '-- inspect-everything philosophy       <- SIMple Physics          ->
    labels, help overlay, STATUS

```

3 Porting recommendations (and what posim actually did)

Constraint recap: posim allows **zero external crates**, **zero unsafe**, **zero warnings**. That immediately rules some things in and some out.

3.1 Recommended and adopted

idea	source	posim realization
Command→status message protocol	PyBullet	Every window action is a small JSON command; the simulator answers with state. Maps 1:1 onto WebSocket text frames — <code>std::net::TcpListener</code> + a hand-rolled RFC 6455 layer (SHA-1 + base64 included, ~200 lines, fully unit-tested).
Engine/renderer decoupling	Rapier	The physics core never draws. The scene server owns a <i>synchronized copy</i> of the system; the browser is just a view.
Geometry once, poses per frame	Rapier	<code>init</code> message carries shapes/sizes once; 30 Hz frame messages carry only <code>[x,y,z,qw,qx,qy,qz]</code> per body.
Run-mode state machine	Rapier RunMode	Stopped / Running / Paused / Reversing enum drives the playback thread.
Reverse via snapshots	Rapier snapshot / restore	A bounded ring buffer (20,000 frames) of cloned system states; SCENE REVERSE replays it backward and auto-pauses at the beginning. Cheap because PhysicalObjectSystem is Clone .
Control-panel UX	OSP	Toolbar: Start/Pause/Stop/Reverse/step/dt — the exact OSP button set plus reverse.
Status readouts	SOFA, OSP	Status bar: mode, t, dt, E, bodies, hidden, history, camera, fps.
Camera conventions	Irrlicht / QGLViewer / Bullet	Left-drag orbit, wheel zoom, right/shift-drag pan, arrows translate — the cross-simulator standard.
Auto-fit camera	Rapier FRAME_SCENE	<code>fit_distance()</code> = <code>max(12, 2.5× farthest entity)</code> .
Layered drawing	PhysicsHub	Grid layer → trails layer → bodies (painter-sorted) → labels.
Settable dt	PyBullet <code>setTimeStep</code>	SCENE SET_TIME_STEP <dt> and the toolbar dt field (validated: positive, finite).
Debug/user events	PyBullet	Window errors and actions become notebook events (§7).

3.2 Considered and deliberately NOT ported

idea		why not
Shared-memory (PyBullet)	transport	Requires raw memory mapping — impossible without unsafe or an external crate. TCP on localhost is fast enough for 30 Hz frames.
Native GUI toolkits (Qt, Irrlicht, Bevy, egui, ggez)		All are external dependencies. The browser is the one universally-available renderer; the page is embedded in the binary with <code>include_str!</code> .
WebGL/GPU rendering		Needs no dependency, but canvas-2D with manual projection is simpler, debuggable, and fast enough for classroom scenes. A WASM/WebGL path remains open (Rapier's route).
Lua scripting (SIMple)		posim already has its own command language; two languages would confuse the audience.
Fragmented/binary Socket frames	Web-	Rejected explicitly by the server: text-only keeps the protocol inspectable by students.

4 The SCENE command family (complete reference)

All keywords are case-insensitive. Numeric arguments are *term-level* expressions: `-5` is negative five, `2*2` works; write `(1+2)` to use a sum. Full grammar in `grammar.pdf` §3.

command	what it does
SCENE CREATE [port]	Start the scene server (OS-assigned port if omitted), open the browser page, show all entities. Second call reports the existing URL.
SCENE CLOSE (alias SCENE DESTROY)	Shut the server down and disconnect every window.
SCENE TRANSLATE dx dy [dz]	Move the camera target by a world-space offset (dz defaults to 0).
SCENE ROTATE dyaw dpitch	Orbit the camera: azimuth and elevation, in degrees (pitch clamps to $\pm 89^\circ$).
SCENE ZOOM IN / SCENE ZOOM OUT	Zoom by a fixed factor ($1.25\times$).
SCENE ZOOM <f>	Zoom by factor f (> 1 zooms in; e.g. <code>scene zoom 0.5</code> zooms out).
SCENE HIDE [n ALL]	Hide object objN , or everything (default ALL).
SCENE SHOW [n ALL]	Show object objN , or everything (default ALL).
SCENE REFRESH	Re-copy the notebook's current system into the window (clears history).
SCENE REDRAW	Re-send the full scene description to every connected window.
SCENE START	Begin (or resume) forward time-stepped evolution.
SCENE STOP	Halt evolution and clear the recorded history.
SCENE PAUSE	Freeze evolution; SCENE START resumes, history kept.
SCENE REVERSE	Play backward in time through the recorded history; pauses automatically at the beginning. Errors if no history exists yet.
SCENE RESET	Re-initialize the playback: every mutable value and the time return to their initial values — the state last synced at SCENE CREATE/SCENE REFRESH — bit-identically; history and the step counter clear and the mode returns to Stopped. SCENE START (or the window's Start button) then re-runs the simulation from the beginning. The toolbar Reset button calls the very same primitive.
SCENE SET_TIME_STEP dt	Set the playback time step (must be positive and finite).
SCENE STATUS	Report URL, connected windows, mode, t, dt, steps, history size, entities, hidden list, camera.
SCENE EVENTS	Drain and print the asynchronous window $\rightarrow$ notebook event queue.

Notes for the curious:

- The playback loop runs on its own thread at ~ 30 ticks/s (33 ms) and advances a **synchronized copy** of your system — your notebook state is never mutated by the window. SCENE REFRESH re-syncs the copy; a future RUN/STEP in the notebook does not disturb a running scene.
- Every forward step goes through `physical_object::integrate` — the same sundials (CVODE/ARKODE) code path as STEP/RUN. There is no second, lesser integrator hiding in the graphics.
- RESET in the notebook keeps the window open and re-syncs it to the now-empty system — including clearing the box wireframe and wall flags if a BOX existed.
- SCENE RESET is **not** the notebook's RESET: the notebook command empties the system itself, while SCENE RESET only rewinds the window's playback copy to the state it was last synced with. The reply spells it out: `scene playback reset to its initial state (t = 0, 1 entity)`; Start runs the simulation again from the beginning.

5 The wire protocol (WebSocket, JSON text frames)

The page connects to `ws://127.0.0.1:<port>/ws`. One JSON document per frame, both directions.

5.1 Simulator $\rightarrow$ window

```
{ "type": "init", "t": 0.0, "dt": 0.01, "mode": "stopped",
  "camera": { "yaw": -60.0, "pitch": 55.0, "dist": 12.0, "target": [0,0,0] },
  "hidden": [],
  "entities": [ { "i": 0, "mass": 256.0, "charge": 0.0, "shape": "sphere", "radius": 0.35 },
                 { "i": 3, "mass": 2.0, "charge": 0.0, "shape": "cuboid",
                   "half_extents": [0.3, 0.2, 0.1] } ] }
```

Sent on connect, on SCENE REFRESH/REDRAW, after hide/show, and after a playback reset (SCENE RESET / the Reset button). Geometry travels **once** (the Rapier idea). Note that BOX <size> / BOX OFF in the notebook do **not** push a new init on their own — box changes reach an open window only via SCENE REFRESH, and the BOX reply reminds you: (`scene window open: SCENE REFRESH shows the box`). The one exception is RESET, which re-syncs the window itself and clears the box wireframe and wall flags automatically.

Each entity carries its shape-specific geometry: a `sphere` its `radius`, a `cuboid` its `half_extents`, a `torus` its `ring_radius` and `tube_radius`, a `disk` its `radius`, a `cylinder` its `radius` and `half_height`, and — new in the dumbbell release — a `dumbbell` its two sphere radii `r1/r2`, its `rod_radius`, and the sphere-centre offsets `z1/z2` along the local axis (the local origin is the composite COM, so the offsets are mass-weighted and unequal). Any entity whose object was registered with `NEW ... AS <name>` also carries `"name"` — the window prefers it over `objN` for the label:

```
{ "i": 6, "mass": 1.0, "charge": 0.0, "shape": "torus", "ring_radius": 1.5, "tube_radius": 0.5 }
{ "i": 9, "mass": 0.6666666666666666, "charge": 0.0, "shape": "disk", "radius": 1.0 }
{ "i": 11, "mass": 2.0, "charge": 0.0, "shape": "cylinder", "radius": 0.5, "half_height": 0.75 }
```

```
{
  "i":0,"mass":0.0,"charge":0.0,"shape":"cuboid","half_extents":[1.0,4.0,4.0],"wall":true
}
{
  "i":0,"mass":3.5,"charge":0.0,"shape":"dumbbell","r1":0.25,"r2":0.25,
  "rod_radius":0.1,"z1":-0.6428571428571429,"z2":0.35714285714285715,
  "name":"dumbell0"}
}
```

Two optional fields describe the rigid bounding box: each of the six static wall slabs carries `"wall":true` on its entity (the page then skips it in the body-drawing pass), and the message top level gains `"box":4.0` — the inner side length — **absent** when no box exists.

```
{
  "type":"frame","t":1.234,"dt":0.002,"mode":"running","steps":617,
  "history":617,"energy":-592.49,
  "p":[0.15,0.0,0.0],"l":[0.4443,0.0,-1.506],"hidden":[],
  "contacts":[{"i":0,"j":1,"point":[0,0,0],"normal":[1,0,0],"impulse":2.0}],
  "bodies":[{"x,y,z,qw,qx,qy,qz", ...}]
}
```

Broadcast every tick (~ 30 Hz) while any window is connected. Bodies are ordered by entity index; quaternions are **w-first** (the posim convention everywhere). `contacts` carries every collision the playback copy resolved in its latest step — `normal` is the unit action–reaction line from body `i` toward body `j` — and the page draws each one as a fading golden arrow through the contact point (arrowhead on the `j` side, ~ 0.6 s persistence; toggle with **Contacts** / **C**). Verified live: a head-on elastic impact broadcasts `normal` `[1,0,0]`, `point` `[0,0,0]`, `impulse` `2.0` — the exact analytic values — and the arrow renders at the projected contact point.

Alongside `energy`, every frame now carries the other two conserved totals, computed on the playback copy every tick: `p` is the **total linear momentum** and `l` the **total angular momentum about the origin** (each `[x,y,z]`). These are exactly the numbers the “conserved quantities” readout (§1.2) displays.

```
{
  "type":"camera","camera":{"yaw":...,"pitch":...,"dist":...,"target":[...]}
}
```

Sent when the *notebook* moves the camera (SCENE TRANSLATE/ROTATE/ZOOM) so every window follows.

5.2 Window → simulator

```
{
  "type":"cmd","action":"start"} // toolbar buttons:
  // start | pause | stop | reverse | reset | step | step_back | set_dt | refresh
  "type":"cmd","action":"set_dt","value":0.005}
  "type":"camera","yaw":-92,"pitch":71,"dist":7.9,"target":[0,0,0]}
  // gesture sync, throttled to 10 Hz, so SCENE STATUS matches the screen
  "type":"event","level":"error","message":"..."} // e.g. page JS errors
  "type":"request_state"} // ask for a fresh init
}
```

Malformed messages never crash the server; they become error events the notebook can read.

6 Under the hood (for the reader who wants the machinery)

```
posim process
```

```

|-- VM thread (your commands)          SCENE ... --+
|-- listener thread (TcpListener, non-blocking poll) | Arc<Mutex<Shared>>
|-- playback thread (33 ms tick) <-----+
|   Running  -> history.push(system.clone()); integrate::step(dt)
|   Reversing -> system = history.pop_back()   (auto-pause at start)
|   broadcast frame JSON to every outbox (mpsc channels)
'-- per-window threads
    reader: parse WS frames -> handle_client_message (0.5 s poll for shutdown)
    writer: outbox receiver -> WS text frames (write half mutex-shared
        with the reader so pings/close never interleave a frame)

```

- **WebSocket layer** (posim/src/scene/ws.rs): SHA-1 (FIPS 180-4) and base64 (RFC 4648) implemented from the standard, plus RFC 6455 framing (text/close/ping/pong; masked client frames; 126/127 length forms). Unit-tested against the official test vectors, including the RFC's own handshake example (dGhlIHNBhXBsZSBub25jZQ== → s3pPLMBiTxaQ9kYGzzhZRbK+x0o=).
- **HTTP layer**: GET / serves the embedded page (include_str!), GET /ws upgrades. Anything else: 404. Bound to 127.0.0.1 only — the window is local by construction.
- **The page** (posim/src/scene/scene.html): one self-contained HTML/CSS/JS file. Canvas-2D renderer with a z-up orbit camera, perspective projection, painter-sorted bodies, radial-gradient spheres, quaternion-rotated shape wireframes, adaptive grid, world axes, 800-point trails. No frameworks, no network fetches.
- **Shape wireframes** — every extended shape is drawn as a small set of loops and lines, all rotated by the body quaternion so spin is visible: a cuboid keeps its 12-edge wireframe; a **torus** is its outer and inner equators, two tube rings (the tube's top and bottom circles) and four tube cross-sections — enough to read the hole and the spin; a **disk** is its rim plus two diameters; a **cylinder** is its two cap rims plus four side lines.
- **The dumbbell renderer** — a **dumbbell** is drawn as one rigid body: two radial-gradient-shaded spheres (radii r1, r2) at their COM offsets z1, z2 rotated by the body quaternion, joined by the rod's **four silhouette lines** (the lines between the sphere centres, offset radially by rod_radius in four directions). Because the offsets are mass-weighted the heavier end sits closer to the drawn position — the spin and the asymmetry are both visible.
- **The conserved-quantities readout** — a permanent labeled overlay (id="hud", top-left under the toolbar) rebuilt on every frame from the frame message's energy, p and l fields: an E line (8 digits) and P/L lines showing the three components plus the |.| magnitude (5 digits). Entity labels, drawn in the body pass, prefer the init entity's "name" (the user's NEW ... AS registration) and fall back to objN.
- **Shared.initial + reset_playback** — the playback state keeps an initial snapshot: the system as last synced from the notebook (set at SCENE CREATE and by every SCENE REFRESH/RESET re-sync). reset_playback restores it as a bit-identical clone — every mutable value and the time return to their initial values — clears the history ring and the step counter, returns the mode to Stopped, and flags a fresh init for every window. Three doors, one primitive: the toolbar **Reset** button, the window's reset command, and the notebook's SCENE RESET all land in the same function.
- **The rigid bounding box** — when the init message carries "box", the page strokes the interior cube as a dashed #5d84a8 wireframe (12 edges, drawn just after the grid, beneath trails

and bodies). The six static wall slabs that implement the box arrive tagged `"wall":true` and are skipped in the body-drawing pass: the dashed wireframe represents them.

7 Asynchronous data in both directions

The old design was strictly synchronous (one reply per request). The scene link is genuinely asynchronous:

Notebook → **window**. Scene commands broadcast immediately; a running playback streams frames whether or not the notebook is busy.

Window → **notebook**. The window pushes errors, data requests and user actions at any time. They are queued (bounded at 1,000) and reach you two ways:

1. **Interactive / script mode** — type SCENE EVENTS:

```
In [7]: scene events
Out[7]: window connected (1 total)
        window action: start
        error: canvas exploded          <- a page JS error, reported home
```

2. **-machine / Jupyter mode** — posim emits *unsolicited* lines `{"event":"scene","message":"..."}` between replies. The Jupyter wrapper kernel runs a background reader thread that routes real replies to the requester and pushes event lines straight to the notebook front end as `[scene] ...` stream output — errors arrive on stderr (red), everything else on stdout. The `{"op":"events"}` request also drains the queue explicitly.

8 How it was verified

Rust tests (52 total, cargo test -workspace). New for the scene subsystem: SHA-1/base64/accept-key official vectors; an HTTP test that the served page contains the toolbar, status bar and all three gesture handlers; a full WebSocket session test (handshake → init → camera sync → start/pause → event queue → frame broadcast); a playback test proving forward-then-reverse lands back *exactly* on the start state; lexer/parser/VM tests for every SCENE command, including the errors (`scene rotate 15` missing an argument, port > 65535, commands before SCENE CREATE).

Wire-protocol test (jupyter/test_protocol.py, stdlib only): 24 checks green, including the whole SCENE family end-to-end over -machine and the events op.

Kernel test (jupyter/test_kernel.py, real ZMQ): 5 checks green.

Real-browser verification (headless Chrome 150 driven over the DevTools protocol, dispatching genuine OS-level input events): 16/16 checks passed —

```
ok scene shows all simulator entities (4)
ok statusbar reports connection          ok toolbar is present
ok ArrowRight translates the view        ok ArrowLeft translates it back
ok ArrowUp/ArrowDown also translate
ok left-drag rotates (yaw -92 -> -124)  ok left-drag rotates (pitch 71 -> 87)
ok drag does not zoom
ok wheel up zooms in (12.00 -> 7.89)    ok wheel down zooms out (7.89 -> 12.00)
ok keyboard + zooms in                  ok keyboard - zooms out
```

ok	toolbar Start begins evolution	ok	toolbar Pause freezes it
ok	toolbar Reverse plays backward (t 0.084 -> 0.022)		

The gestures were then **independently re-verified** in a second, different browser session (2026-07-22, driving genuine key, drag and wheel events against a live 4-entity scene on port 7878): ArrowRight moved the camera target along the view's right axis and ArrowLeft returned it **exactly** to $[0, 0, 0]$; a left-drag changed yaw $-60^\circ \rightarrow -98.8^\circ$ and pitch $55^\circ \rightarrow 32.2^\circ$ while distance and target stayed untouched (a pure orbit); wheel-up/down zoomed by the exact 1.15 factors ($12 \rightarrow 10.435 \rightarrow 12$) and the +/- keys by the exact 1.2 factors ($12 \rightarrow 10 \rightarrow 12$); toolbar Start advanced t through the CVODE playback path, Pause froze it, and Reverse replayed history backward until it auto-paused at **exactly** $t = 0$ with the initial energy $E = -194.84952$ restored to the last digit.

The four self-checking physics examples (Kepler, outer solar system, tumbling body, charged particle) still print SUCCESS, the build is warning-free, and Cargo.lock still lists exactly the five local crates — the zero-dependency proof.

Shapes-and-box release (2026-07-24). Two further layers of checks accompany the TORUS/DISK/CYLINDER shapes and the BOX command (the workspace suite is now 94 tests: 37 lib + 15 collision + 9 conservation + 33 posim, zero warnings):

- **The init message carries the new geometry.** The new posim test `box_shapes_and_wall_flags_reach` opens a genuine WebSocket to the scene server and asserts the init JSON contains `"box":4.0`, `"shape":"torus"` with `"ring_radius":1.5` and `"tube_radius":0.5`, and `"wall":true` on the static slab.
- **Live browser session.** The 12-entity demo (`scripts/collisions/11_box_of_shapes.posim`: six walls plus a torus, point, sphere, disk, cube and cylinder inside BOX 4) was played back in a real browser window: the window's playback copy conserved energy to $\sim 1\text{e-}9$; a pixel scan of the canvas found the dashed box wireframe (5,904 pixels of #5d84a8) and the golden contact arrows (2,218 pixels); toolbar Start and Reverse were exercised through real DOM clicks; the JS console stayed free of errors.

Reset + conserved-quantities release (2026-07-24). The workspace suite is now **99 tests: 39 lib + 16 collision + 9 conservation + 35 posim**, zero warnings. What the new layers pin down:

- **Reset is a true re-initialization.** The posim test `reset_restores_the_initial_state_and_start_reru` runs the playback forward (≥ 10 steps), calls the reset primitive, and asserts the mode is Stopped, the time is back to its initial value, the packed 13-per-object state is **bit-identical** to the initial pack, and the step counter is 0 — then sets Running again and asserts the simulation re-starts. The page-serve test additionally asserts the permanent `id="bt-reset"` button is in the served HTML, and `scene_commands_compile` covers SCENE RESET.
- **Reset verified live.** In a real browser session the demo was run to $t = 0.72$, the toolbar Reset was clicked: mode `stopped`, $t = 0$, history = 0, and the body's state bit-identically back at its start values; clicking Start then set the window running again from the beginning.
- **The protocol carries the new fields.** The WebSocket session test (`websocket_session_end_to_end`) asserts every broadcast frame contains `"p":[` and `"l":[`; `box_shapes_and_wall_flags_reach_the_init_me` asserts the init JSON carries the registered user name (`"name":"ringo"`).

- **The dumbbell demo, live.** The two-dumbbell collision demo (`scripts/collisions/12_two_dumbbells.py`) — two user-named dumbbells built by a user-defined function, colliding off-center with spin) was played back in a real browser window: the entity labels read `dumbell0` and `dumbell1`, and the conserved-quantities readout showed

```
E = 7.86531061, P = [0.15000, 0.00000, 0.00000] |.| = 0.15000, L = [0.44430,
0.00000, -1.50600] |.| = 1.57017
```

identically before and after the impact (L's y component showed `-1.31228e-13` after — machine-epsilon noise). The notebook run of the same script reports, through 2 real CVODE collision events,

```
t = 3 (3861 solver steps, 60 snapshots, |dE/E| = 9.463e-11, 2 collision(s) --
CONTACTS lists them)
```

with the total momentum `[0.150000000000000036, 0, 0]` **bit-identical** before and after.

9 FAQ / troubleshooting

No window appeared after SCENE CREATE.

The command prints the URL (e.g. `http://127.0.0.1:41234/`). Open it in any browser yourself — the automatic opening uses `xdg-open`, which headless machines lack.

I want a predictable port.

`SCENE CREATE 7878` → the URL is always `http://127.0.0.1:7878/`. If the port is busy you get a clear error; pick another or use 0 (OS-assigned).

Stop the browser from opening (tests, servers).

Set the environment variable `POSIM_NO_BROWSER=1`.

REVERSE says there is nothing to reverse.

Reverse replays *recorded* history. Run `SCENE START` (or single steps) first; `STOP` clears the recording, `PAUSE` keeps it.

The window shows old objects.

The window views a synchronized *copy*. After creating/deleting objects in the notebook, type `SCENE REFRESH`.

Is any of this sent over the internet?

No. The server binds `127.0.0.1` (loopback) only.

Where do error messages from the window go?

Into the event queue: `SCENE EVENTS` in the notebook, `[scene] ...` lines in Jupyter, `{"event":...}` lines in `-machine` mode.